

POWER TRANSFORMERS Used In Electronic Products

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This project report is focussed on setting up a production unit for step-down power transformers that are generally used in consumer electronics

Transformer, an electromechanical component, transfers electrical energy from one circuit to another by means of changing the magnetic field, without changing frequency. Various types of transformers are used for different electronic applications. Depending on applications, transformers can be classified into the following broad categories:

1. Power transformer
2. Current transformer
3. Constant-voltage transformer
4. Impedance-matching transformer
5. Driver transformer
6. Audio-output transformer
7. Ignition transformer
8. Pulse transformer
9. Fly-back transformer
10. Ferrite-core transformer
11. Intermediate-frequency (IF) transformer

The main difference between the various types of transformers is the type of core used and the number of turns in primary and secondary coils. Advancements in the field of electronics have influenced change in design and use of new type of core materials. For example, ferrite-core transformers are used in switch-mode power supplies and electronic chokes, dynamo-type core to silicon-grade cold-rolled steels for general-type nickel iron alloys, ferrite-core for high quality and high reliability, and so on. This project report is focussed on step-down power transformers that are generally used in consumer electronics.

Business compliances

In the beginning, as a manufacturer apply for different registrations and licences from the relevant government authorities. However, it is advisable to check the state law.

1. Register the business with Registrar of Companies (ROC).
2. Apply for trade licence from local municipal authority.
3. Apply for MSME Udyog Aadhaar for small- and medium-unit operations. The same can be registered online.
4. Obtain NOC from Pollution Control Board. First, apply for Consent to Establish and then for Consent to Operate.
5. Apply for GST Registration.
6. Protect brand name with trademark registration.

Market potential

Transformers and coils are used in almost all electronic products, including consumer electronics, process control instruments, power electronics, medical electronics, industrial electronics, defence electronics and telecommunications, to perform various functions. Performance of an electronic equipment depends on the quality of transformer. Therefore growth of the electronics manufacturing industry also drives the demand for transformers.

There is huge scope for the transformer and coil industry if high

quality and competitive costs are maintained. Moreover, it is relatively easy to start a transformer manufacturing business, as manufacturing equipment and raw materials are indigenously available. However, as a new entrepreneur you must manufacture different types of transformers, and have marketing tie-ups with electronics manufacturing companies to ensure economic viability.

Manufacturing process

The manufacturing process starts with quality checks. Incoming raw materials must be tested for required quality before being released for production. Then, super-enamelled copper wires are wound on bobbins or former, as per specifications, using a CNC winding machine. Coils are then stalked with core material, terminated and clamped. The completed transformer and coils are vacuum-impregnated using varnish and baked in an oven at a particular temperature.

Important manufacturing steps are explained below.

Core forming. High-quality cold-rolled grain oriented (CRGO) laminations are used in transformers. These come in the form of thin sheets and are cut as per the design, varying for different capacity transformers. Highly skilled people assemble the laminations on a core channel to form the core. These should be perfectly assembled without any gaps to avoid energy loss.

Winding. Both high and low voltage windings are done using coil winding machines. Copper or aluminium strips/wires used in winding are meticulously selected for their quality to give the best output. These strips/wires are paper-covered and act as insulators. This helps increase short-circuit strength, thermal strength and maintain higher efficiency.

Assembling. Wound coils are placed carefully in the assembled core. Insulations are given wherever required using materials like press board. The core bolt and tie rods are fixed in position. Primary and secondary windings are connected as per requirement. Complete assembly is kept in the hot air chamber, and a high temperature is maintained to prevent any moisture in the core coil assembly.